

QUALITATIVE STUDY · SERVICE-DOMINANT LOGIC · IMMERSIVE VR

From Immersion to Attenuation

Value Co-Creation and its Decline in a Social Virtual World

Interaction activities → value

Co-creator roles as trajectory

Co-creation **attenuation**

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“At the beginning everyone was very nice... now it's all map bombers and people swearing...”

I'll play one last day and then never come back.”

– I10, A USER WHO HAD BEEN AMONG THE MOST COMMITTED, NOW LEAVING

The **same engagement** that builds a community's value can, over time, produce the conditions for its decline — and its most invested members can become its most consequential critics.

Co-creation is theorised as a one-way climb. Decline is treated as a separate accident.

The optimistic account

Service-dominant logic casts users as active co-creators; online-community research assumes deeper participation **cumulatively** enhances value, drawing members from periphery to centre.

The negative side — but fragmented

Value co-destruction, the dark side of engagement, fatigue & platform decline each capture decline as **discrete failures** or individual withdrawal — detached from the value-creating process they emerge from.

The gap: how creation and erosion are linked within a *single trajectory of participation* — and how immersive, embodied VR intensifies both.

- **RQ1 · Activities → value.** Through what interaction activities do users co-create value in an immersive social virtual world, and what **forms of value** do these activities generate?
- **RQ2 · Roles.** How do these activities configure different users into distinct **co-creator roles**?
- **RQ3 · Attenuation.** How do **structural barriers** progressively attenuate co-creation across these roles and over time?

The logic is cumulative: **activities create value → participation crystallises into roles → barriers withdraw the conditions that sustain them.**

Value is not embedded in outputs but **co-created in use**, when actors integrate resources through platform affordances (Vargo & Lusch, 2004, 2008).

We move beyond "users are resource integrators" to ask the compositional question:

which resources, through which affordances, by which actors, generate which forms of value?

SOCIOTECHNICAL SPINE

User resources — skill, creativity, social ties, emotional labour, time, attention — meet **platform resources** — spatial infrastructure, creation tools, avatar systems, real-time voice, governance.

Co-creation lives at the **intersection of the social and the technical** — neither alone.

ABDUCTIVE QUALITATIVE DESIGN

- Iterating between data and theory (Dubois & Gadde, 2002) — categories emerge from data, refined against co-creation theory.
- Two sources **triangulated**: immersive netnography + interviews.

LIGHT WORLD – WHY THIS SITE

- **VR-based SVW** on PICO headsets: avatar-mediated, user-built persistent worlds, spatial voice; ~10-16 co-present per world.
- Chosen as a **revelatory, not representative**, case (Patton, 2015): it made the **full arc — including decline** — observable in one setting.

A site selected precisely because it lets us watch co-creation **both build and erode** — the aim was a revelatory case, not a typical one.

~100h

immersive netnography, Jan–Jun 2026, across days & times

- **Abductive, multi-stage coding** (Gioia et al., 2013): first-order codes → second-order themes → aggregate dimensions.
- Built **from the data upward**; coded to **thematic saturation**.

457

public chat logs · 110 screenshots · 8,000-word reflexive diary

10

semi-structured interviews, 30–45 min, verbatim

- First author coded; second **independently cross-checked**; team reviewed the data structure.
- **Audit trail & reflexivity** — including the observer effect of a persistent avatar.

5 → exit

sampled across the arc: newcomers · highly engaged · lapsed & exited

FINDING 1 • RQ1 FOUR INTERACTION CLUSTERS, FOUR FORMS OF VALUE

Exploratory & experiential wandering, browsing, dwelling	→	Experiential & aesthetic value
Creative & collaborative building worlds, props, avatars	→	Content & symbolic value
Social & emotional friendships, gatherings, support	→	Relational & emotional value
Governance & negotiation reporting, demanding, negotiating	→	Normative & institutional value

“no matter how big your imagination, you can realise it with the editor”

– I09, ACTIVE CREATOR

“I made a lot of friends here, and they made this app super fun”

– I01, ORGANISER

01 • peripheral

Experiential participant

Supplies attention & presence — the audience base.

02 • moving in

Creation-oriented prosumer

Technical & design skill — builds the content base.

03 • central

Social organiser

Social capital + **emotional labour** — traffic & cohesion.

04 • central

Knowledge mentor

Expertise + emotional labour — lowers entry barriers.

♦ PEAK

05 • central → exit

Critic / negotiator

Scrutiny + prior investment — **signals governance failure.**

Roles are **fluid** and **functional**: not fixed types, but a division of labour users move through — a **layered interaction architecture**, not a static typology.

ENGAGEMENT IS VALUE AND VULNERABILITY

The critic role is typically the **destination** of the trajectory — not an outside disruptor.

Many critics **began as committed creators and organisers**, and turned only after repeated, unaddressed barriers. Because they invested most, they are the **most credible critics** when governance fails — and their disillusioned discourse colours newcomers and mobilises others toward exit.

Two sub-mechanisms

- 1 • Committed → critic conversion.** The most invested don't leave quietly; they reshape the community's reputation.
- 2 • Emotional-labour exhaustion.** The unpaid warmth of organisers & mentors tips into burnout once technical & governance support falls away.

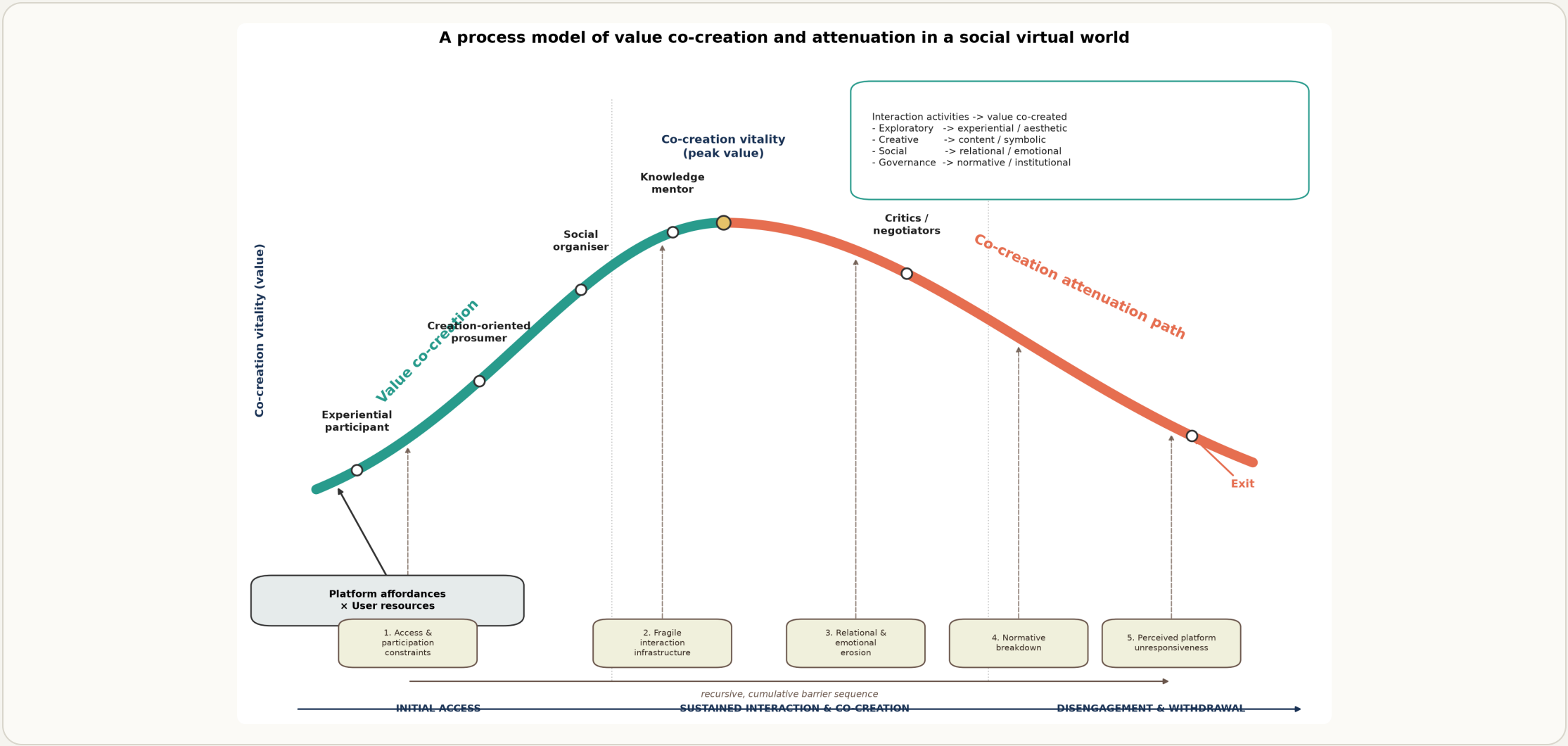
This overturns a common assumption: the most engaged users are *not* unconditionally beneficial.

FINDING 3 • RQ3 FIVE STRUCTURAL BARRIERS — A RECURSIVE SEQUENCE, NOT A LIST

- 1 **Access & participation constraints** — VR-only reach + a complex toolkit narrow the creator base at the source.
- 2 **Fragile interaction infrastructure** — lag, crashes, logic failures make creative labour feel wasted.
- 3 **Relational & emotional erosion** — thinning content drives audiences away; emotional labour turns to exhaustion.
- 4 **Normative breakdown** — thinned, weakly-governed spaces fill with map bombing & abuse.
- 5 **Perceived platform unresponsiveness** — official silence converts voice into exit, and private grievance into public criticism.

Recursive, not parallel: each barrier seeds the next — a *sociotechnical cascade* from a technical fault to social erosion to governance failure. We name this cumulative erosion **co-creation attenuation**.

THE MODEL ONE TRAJECTORY: RISE, PEAK, ATTENUATION



CONTRIBUTION CO-CREATION ATTENUATION ≠ VALUE CO-DESTRUCTION

CONCEPT	LEVEL	TEMPORALITY	MECHANISM
Value co-destruction	Dyadic (actor-actor)	Discrete, within an encounter	Resource misintegration
Dark side of engagement	Individual / relational	Discrete or recurrent	Over-engagement, conflict
Fatigue / overload	Individual	Gradual psychological state	Exhaustion, reduced motivation
Disengagement	Individual	Downstream exit	Withdrawal, dissatisfaction
Co-creation attenuation (this study)	Ecosystem / community	Longitudinal, cumulative	Recursive erosion of structural conditions

1 · Embodied co-creation

Extends value co-creation into place-based, avatar-mediated VR — embodiment & spatial attachment as resources.

2 · Roles as architecture

Reframes co-creator roles from static typology into a functional architecture along an engagement trajectory.

3 · Attenuation

A processual, ecosystem-level construct — antecedent to, and softer precursor of, co-destruction.

- **Read the early markers.** Creators quitting, public spaces emptying, reports going unanswered — treat attenuation as detectable and manageable.
- **Responsiveness is the highest-leverage lever.** Because unresponsiveness closes the recursive loop, visible action forestalls the committed-to-critic conversion.
- **Protect the emotional-labour roles.** Organisers & mentors hold communities together through unpaid labour — shielding them from burnout is both ethical and strategic.
- **Govern role by role.** Different roles need different support — from stable cross-device creation tools to transparent, visible feedback loops.

The practical question is not how to transact, but how to *sustain co-creation over time* — a continuous, sociotechnical task across access, infrastructure, relationships and governance.

A community does not simply accumulate value or, separately, suffer failure. It builds co-creative vitality — and then loses it along the very trajectory that created it.

THREE QUESTIONS FOR THE DISCUSSION

Framing the core construct

Does "co-creation attenuation" land most powerfully as a **sociotechnical** process — or should S-D logic stay in the foreground?

Evidencing the pivot

How far can the **committed-to-critic** pathway be pushed on cross-sectional data before it needs a longitudinal design?

Target & positioning

Is this best positioned for an **information-systems** audience, and where would you push the theorising further?

Thank you — **I'd welcome your feedback.** · Qingyun Lyu